M10 Series RGB-D Cameras

V2.0 2026-10-29

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Helping Robots Understand the World



MRDVS M10 series industrial RGB-D cameras combine compact size, high performance and cost efficiency. With 480 × 320 resolution, they achieve centimeter-level accuracy in bag/box depalletizing and volume measurement, and simultaneously output high dynamic range RGB, depth and IR images plus 3D point clouds. dToF provides higher-quality data than iToF on high- and low-reflectivity surfaces. M10-CP offers a wide FoV optimized for close range.

Features

High Resolution and Accuracy

dToF and 480 × 320 resolution deliver accurate, stable ranging in complex indoor/outdoor dynamic scenes, with a wider effective detection range.

Strong Interference Immunity

Compared with iToF, the M10 series better handles overlapping reflected signals and interference, with greater sunlight immunity for reliable range data.

Synchronized Data Outputs

Simultaneously outputs high-precision RGB data, depth and IR images, and 3D point clouds, with no acquisition delay.

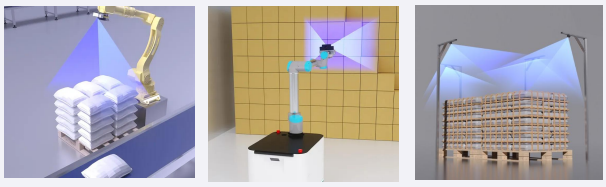
Applications

1 Bag/Box Depalletizing

RGB images and point clouds, combined with AI, locate targets and their orientation to guide robot arms or end effectors (suction cups or grippers) for picking randomly arranged objects.

2 Volume Measurement

The M10 series captures 3D object data in complex environments to calculate volume accurately, for logistics, warehousing and automated production lines.



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Specifications

Model	M10	M10-CP (Close-range Optimized)
Sensor Type	dToF RGB-D Camera	dToF RGB-D Camera
Laser Wavelength	940 nm	940 nm
Output Formats	Depth / IR / RGB / 3D point clouds	Depth / IR / RGB / 3D point clouds
ToF Resolution / Frame Rate	480×320@5fps	480×320@5fps
ToF FoV	90°×60°[1]	**120°×80°**[3]
RGB Resolution / Frame Rate	1920×1080@5fps/max20fps	**1632×1224@5fps/max10fps**
RGB FoV (H × V)	90°×60°[2]	**120°×90°**[4]
Range	0.5-5 m @ 10% reflectivity	**0.1-3.5 m @ 10% reflectivity**
Accuracy	±30mm	**±20mm**
Average Power Consumption	≤4W	≤4W
Dimensions	80mm×37mm×25mm	80mm×37mm×25mm
Weight	Approx. 190 g	Approx. 190 g
Power Supply	12V~27V	12V~27V
Communication Interface	Ethernet	Ethernet
IP Rating	IP54	IP54
Eye Safety	CLASS I	CLASS I
Operating Temperature	-20°C~60°C	-20°C~60°C
Storage Temperature	-40°C~70°C	-40°C~70°C
Time Synchronization	PTP	PTP
Software Environment	C/C++/ROS SDK	C/C++/ROS SDK
Supported Operating Systems	Windows 7/8/10/11, Linux, Arm Linux/ROS	Windows 7/8/10/11, Linux, Arm Linux/ROS
Ambient Light Immunity	100 kLux	100 kLux

Notes [1]-[4]: Field of view (FoV) tolerance is up to ±3°.

